[fn20] A Hilbert space.

[fn21] These are the eigenvalues of the operator associated with the physical variable in question. The key equation is the eigenvalue equation.

[fn22] This cloud is described by a mathematical object called wave function. The Austrian physicist Erwin Schrödinger has written an equation describing its evolution in time. Quantum mechanics is often mistakenly identified with this equation. Schrödinger had hopes that the 'wave' could be used to explain the oddities of quantum theory: from those of the sea to electromagnetic ones, waves are something we understand well. Even today, some physicists try to understand quantum mechanics by thinking that reality is the Schrödinger wave. But Heisenberg and Dirac understood at once that this would not do. To view Schrödinger's wave as something real is to give it too much weight – it doesn't help us to understand the theory; on the contrary, it leads to greater confusion. Except for special cases, the Schrödinger wave is not in physical space, and this divests it of all its intuitive character. But the main reason why Schrödinger's wave is a bad image of reality is the fact that, when a particle collides with something else, it is always at a point: it is never spread out in space like a wave. If we conceive an electron as a wave, we get in trouble explaining how this wave instantly concentrates to a point at each collision. Schrödinger's wave is not a useful representation of reality: it is an aid to calculation which permits us to predict with some degree of precision where the electron will reappear. The reality of the electron is not a wave: it is how it manifests itself in interactions, like the man who appeared in the pools of lamplight while the young Heisenberg wandered pensively in the Copenhagen night.

[fn23] Dirac's equation.

[fn24] There is a phenomenon which seems not to be reducible to the standard model: 'dark matter'. Astrophysicists and cosmologists observe in the universe effects of matter which seems not to be the type of matter described by the standard model. Out there, there are still many things that we don't know.

[fn25] I find the claim that the Higgs boson 'explains mass' exaggerated. The Higgs boson does not 'explain' anything about the origin of mass. What would 'explain' the mass of the Higgs? The point is technical: the standard model relies on certain symmetries, and these symmetries seemed to permit only particles devoid of mass. But Higgs and others realized that it is possible to have both symmetries and mass, as long as the latter enters indirectly via the interaction with the field known today as the Higgs field.

[fn26] A finite region of the phase space – the space of the possible states of a system – contains an *infinite* number of distinguishable classic states, but *always* only a *finite* number of orthogonal quantum states. This number is